

Phylogenetically-Informed Leaf Physiognomy: Methodology and Results

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1 Introduction

This document describes the methodology and current results of a project predicting paleoclimate from leaf morphology using phylogenetically-informed prediction (PIP). The primary goal is predicting mean annual temperature (MAT) and mean annual precipitation (MAP) from the morphological traits of fossil leaves. The extant-species modelling pipeline serves as a calibration and validation benchmark.

The dataset is the Royer et al. (2012) leaf morphology and climate database, matched to a dated angiosperm phylogeny derived from the World Checklist of Vascular Plants (WCVF). All analysis is conducted at the species level using species-mean trait values.

Two climate targets are modelled:

- **MAT**: mean annual temperature (°C)
- **log(MAP)**: natural log of mean annual precipitation (mm/yr), back-transformed for reporting

2 Data

2.1 Trait selection

All models are restricted to traits that can be measured from fossil leaf specimens. This ensures the training and prediction pipelines use the same feature set. Training on traits that cannot be measured from fossils would make the model inapplicable to the fossil case.

The 12 fossil-measurable predictors are:

Trait	Notes
Always measurable	
ln.leaf.area.mm2	Log leaf area (ln mm ²)

feret.diam.ratio	Ratio of perpendicular diameters
pw2.a.ratio	Petiole width ² / area
margin.score	0 = toothed, 1 = untoothed, 0.5 = mixed
Toothed leaves (filled for untoothed)	
perim.ratio	Set to 1 for untoothed leaves
teeth.perimeter.percm	Set to 0 for untoothed leaves
teeth.interior.percm	Set to 0 for untoothed leaves
avt.tooth.area	Set to 0 for untoothed leaves
tooth.area.blade.area.ratio	Set to 0 for untoothed leaves
tooth.area.perimeter	Set to 0 for untoothed leaves
tooth.area.interior	Set to 0 for untoothed leaves
teeth.blade.area.ratio	Set to 0 for untoothed leaves

Excluded traits include **evergreen** (phenological status, not determinable from a fossil), raw tooth counts, and several size and shape measurements that require intact leaf material not always preserved in fossils.

2.2 Tooth trait filling

Untoothed leaves have blank tooth trait values in the dataset. For leaves confirmed as untoothed (**margin.score** == 1), these blank values are biologically meaningful zeros, not missing data. They are filled before species-level aggregation:

- Tooth traits: set to **0**
- **perim.ratio**: set to **1** (smooth perimeter)

Both conditions must be met: the cell is blank and **margin.score** == 1. Without this step, tooth traits have roughly 67% missing values at the specimen level and are dropped from all models.

2.3 Species-level and site-level aggregation

The phylogenetic model operates at the species level, so the primary dataset uses species-mean trait values. After aggregation, it contains **1847 species** with at least partial trait information.

A second dataset aggregates all specimens within each site to site-mean trait values, giving **92 sites**. This is the aggregation scale used by traditional leaf physiognomy methods (CLAMP, DiLP), where all specimens from a single site are treated as one observation. Site-level models are covered in Section 6.

3 Non-Phylogenetic Baseline Models

Three non-phylogenetic model types are fit at the species level: ordinary least squares linear regression (LM), regularised linear regression (Elastic Net), and Random Forest. These models provide a reference for evaluating PIP. Site-level models are covered in Section 6.

3.1 Model setup

All models are fit using 10-fold cross-validation via the `caret` framework, predicting MAT and log(MAP) separately. Predictors are centred and scaled. Missing values are imputed once before cross-validation using bootstrap-aggregated regression trees (`bagImpute`). Imputing within each fold would repeat the procedure 60 times (10 folds $\times$ 3 methods $\times$ 2 targets) with little benefit.

For Elastic Net, the mixing parameter and regularisation penalty are selected over a grid of 10 candidate values. For Random Forest (via `ranger`), variable importance uses the impurity criterion, computed during tree construction.

3.2 Cross-validation results

Ten-fold CV RMSE across folds for each model and target:

Table 2: MAT CV RMSE ($^{\circ}\text{C}$), distribution across 10 folds

	Min.	X1st.Qu.	Median	Mean	X3rd.Qu.	Max.	NA.s
LM	5.089	5.347	5.540	5.506	5.719	5.842	0
ENet	5.253	5.296	5.445	5.476	5.600	5.804	0
RF	4.649	5.086	5.121	5.182	5.247	5.742	0

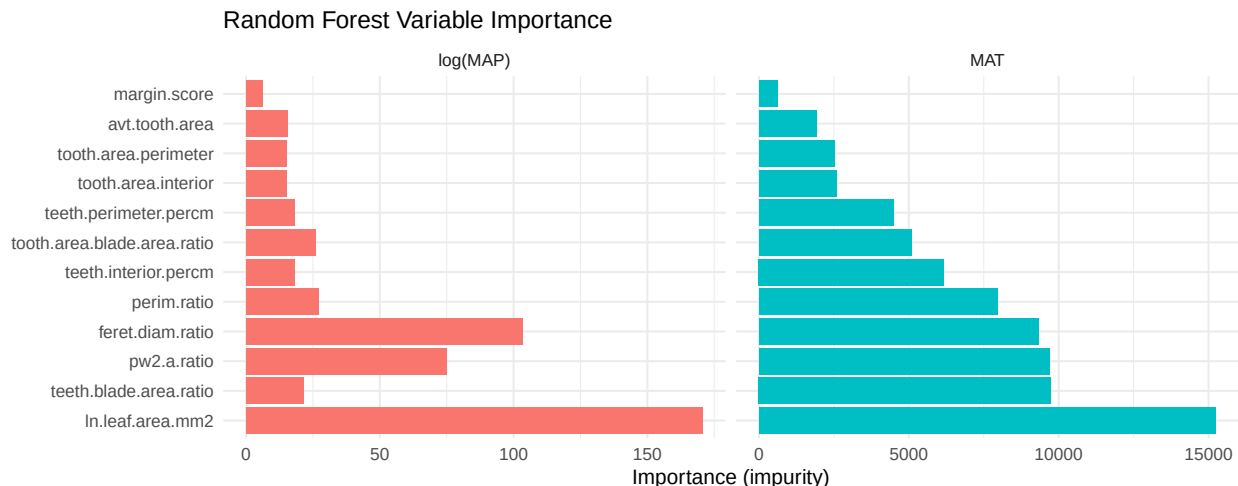
Table 3: log(MAP) CV RMSE, distribution across 10 folds

	Min.	X1st.Qu.	Median	Mean	X3rd.Qu.	Max.	NA.s
LM	0.514	0.550	0.561	0.577	0.574	0.744	0
ENet	0.508	0.539	0.567	0.558	0.580	0.598	0
RF	0.489	0.512	0.522	0.531	0.543	0.612	0

Random Forest has the lowest mean CV RMSE among the non-phylogenetic methods for both targets.

3.3 Variable importance

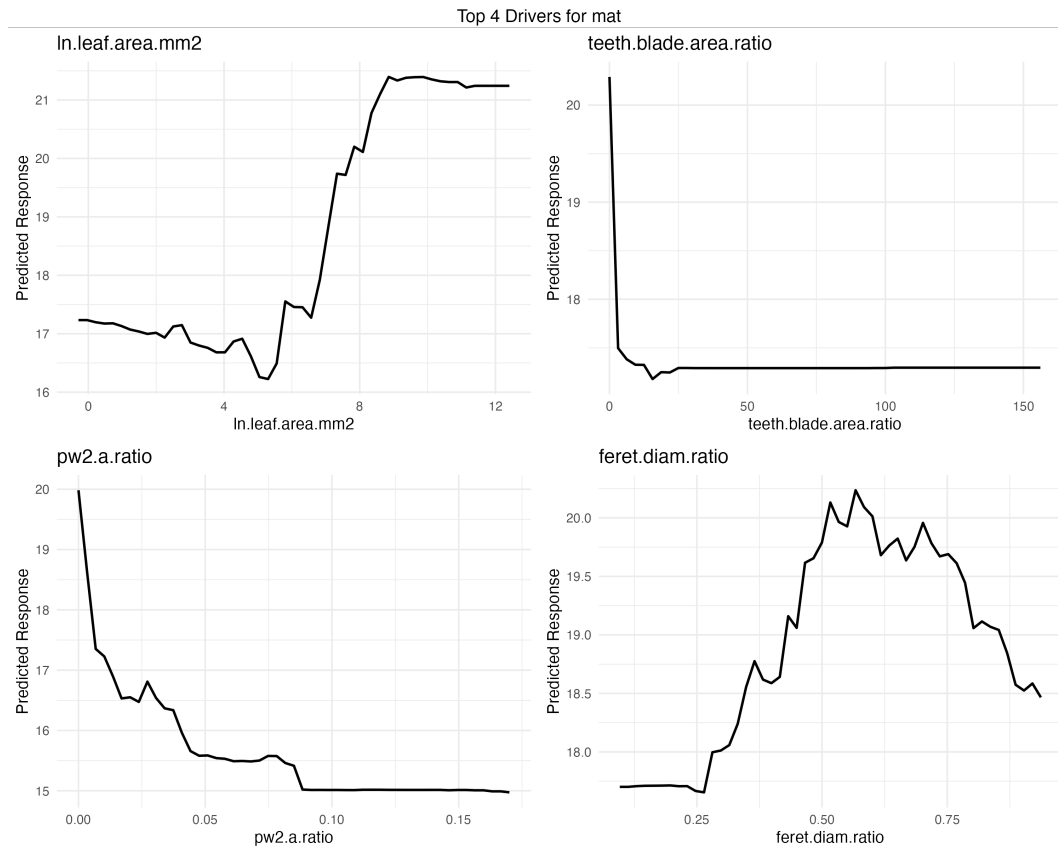
Predictor importance from the Random Forest models (impurity-based, unscaled):



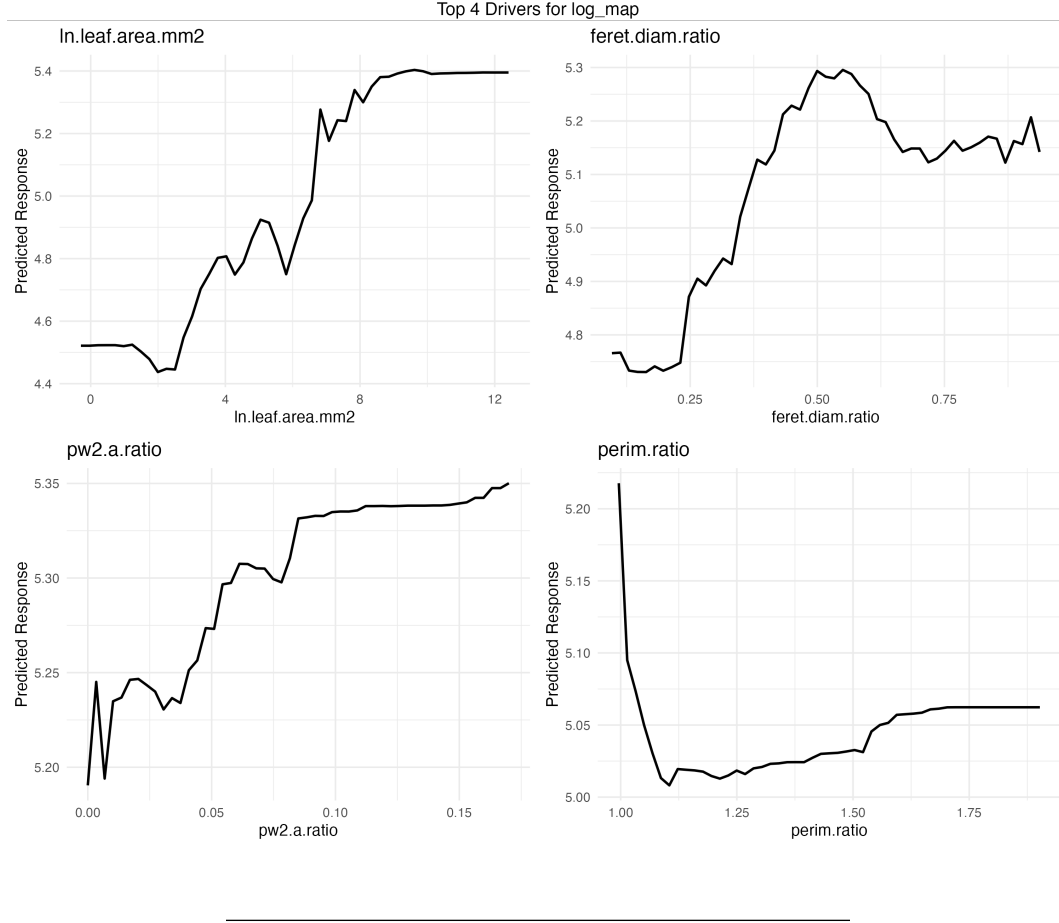
3.4 Partial dependence plots

Each plot shows the marginal effect of a predictor on the model response, averaged over the distribution of all other predictors.

MAT: top 4 predictors



log(MAP): top 4 predictors



4 Phylogenetically-Informed Prediction

4.1 PGLS and Pagel's lambda

Species share traits through common ancestry. Closely related species tend to have similar trait values, making them non-independent observations. Phylogenetic generalised least squares (PGLS) accounts for this by modelling the residual covariance using the phylogenetic variance-covariance (VCV) matrix.

The VCV matrix $\mathbf{V}$ is derived from the dated angiosperm phylogeny. Entry $V[i,j]$ equals the height of the most recent common ancestor (MRCA) of species i and j , which is the total shared evolutionary time between them. The diagonal $V[i,i]$ is the full root-to-tip distance for species i .

Pagel's lambda (λ) is estimated by maximum likelihood. It scales the off-diagonal elements of $\mathbf{V}$ before fitting, while leaving the diagonal unchanged:

$$\mathbf{V}_\lambda[i,j] = \lambda \cdot \mathbf{V}[i,j] \quad (i \neq j), \quad \mathbf{V}_\lambda[i,i] = \mathbf{V}[i,i]$$

- $\lambda = 1$: residuals follow the phylogenetic expectation (Brownian motion)
- $\lambda = 0$: residuals are independent of phylogeny, equivalent to OLS
- $0 < \lambda < 1$: intermediate case

ML-estimated lambda values:

Target	
MAT	0.657
log(MAP)	0.483

Both values are greater than zero, indicating some phylogenetic structure in the residuals for both targets.

4.2 Active predictors selected by Elastic Net

All 12 fossil-measurable traits are used to fit the non-phylogenetic models. For PGLS, the active predictor set is reduced to the traits with non-zero Elastic Net coefficients. This reduces the risk of overfitting and multicollinearity in the PGLS model.

Target	Active trait
MAT	pw2.a.ratio
	ln.leaf.area.mm2
	feret.diam.ratio
	margin.score
	perim.ratio
	teeth.perimeter.percm
	teeth.interior.percm
	avt.tooth.area
	tooth.area.interior
	teeth.blade.area.ratio
log(MAP)	pw2.a.ratio
	ln.leaf.area.mm2
	tooth.area.blade.area.ratio
	tooth.area.perimeter

4.3 The PIP prediction formula

PIP extends PGLS to predict climate values for new species (fossils) using the phylogenetic covariance between the fossil and the training set. Define:

- $\mathbf{y}$: observed climate values for n extant training species
- $\mathbf{X}$: design matrix of trait predictors for training species
- β : PGLS coefficient vector
- $\mathbf{e} = \mathbf{y} - \mathbf{X}\beta$: PGLS residuals from the full-data fit
- $\mathbf{V}[\text{train}, \text{train}]$: -transformed VCV among training species
- $\mathbf{V}[\text{fossil}, \text{train}]$: cross-covariance between each fossil and all training species, scaled by

The prediction for a fossil specimen is:

$$\hat{y}_{\text{fossil}} = \mathbf{X}_{\text{fossil}}\beta + \mathbf{V}[\text{fossil}, \text{train}] \cdot \mathbf{V}[\text{train}, \text{train}]^{-1} \cdot \mathbf{e}$$

The first term is the regression prediction from the fossil’s leaf traits. The second term is the phylogenetic adjustment. It is a weighted sum of the training-species residuals. The weights are proportional to shared evolutionary history between the fossil and each training species. A fossil with no close relatives in the training set receives a near-zero adjustment, and its prediction depends almost entirely on the first term.

The implementation follows the source code of Gardner, Baker, Venditti & Organ (2024; *Nature Communications*), originally from Freckleton (2015), via the `weights.p()` and `pglmPredictMissing()` functions.

4.4 Matrix algebra detail (*optional*)

4.4.1 GLS coefficient estimation

Given the λ -transformed VCV $\mathbf{V}_\lambda$, PGLS estimates coefficients by generalised least squares:

$$\beta = (\mathbf{X}^\top \mathbf{V}_\lambda^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}_\lambda^{-1} \mathbf{y}$$

The coefficient covariance matrix is:

$$\Sigma_\beta = \hat{\sigma}^2 (\mathbf{X}^\top \mathbf{V}_\lambda^{-1} \mathbf{X})^{-1}$$

where $\hat{\sigma}^2 = \varepsilon^\top \mathbf{V}_\lambda^{-1} \varepsilon / (n - k)$ is the GLS residual variance.

4.4.2 Cross-covariance block

The cross-covariance $\mathbf{V}[\text{fossil}, \text{train}]$ is extracted from the VCV of the combined tree (training species plus fossils), restricted to the off-diagonal block:

$$\mathbf{V}[\text{fossil}, \text{train}]_{ij} = \lambda \cdot h(\text{MRCA}(\text{fossil}_i, \text{train}_j))$$

where $h(\text{MRCA})$ is the height of the most recent common ancestor in the time-calibrated tree. This block uses `only` only, with no diagonal correction, because all entries are cross-species (off-diagonal) covariances.

4.4.3 Prediction standard error

The reported standard error propagates coefficient uncertainty through the fossil design matrix:

$$\text{SE}(\hat{y}_{\text{fossil}}) = \sqrt{\mathbf{x}_{\text{fossil}}^\top \Sigma_\beta \mathbf{x}_{\text{fossil}}}$$

This does not include the conditional phylogenetic variance:

$$\text{Var}_{\text{phylo}} = \text{diag}(\mathbf{V}[\text{fossil}, \text{fossil}] - \mathbf{V}[\text{fossil}, \text{train}] \cdot \mathbf{V}[\text{train}, \text{train}]^{-1} \cdot \mathbf{V}[\text{train}, \text{fossil}]) \cdot \hat{\sigma}^2$$

or imputation uncertainty for missing traits. The reported 95% CIs are therefore narrower than full prediction intervals.

4.5 Leave-one-out cross-validation

PIP LOOCV uses the Schur complement identity to avoid refitting the model N times. Let $\mathbf{K} = \mathbf{V}_{-}^{-1}$. The leave-one-out prediction for species i is:

$$\hat{y}_{-i} = y_i - \frac{(\mathbf{K}\varepsilon)_i}{K_{ii}}$$

This follows from the PIP formula applied leave-one-out, using the block-matrix identity $\mathbf{V}[i, -i] \cdot \mathbf{V}[-i, -i]^{-1} = -\mathbf{K}[i, -i] / K_{ii}$. The result is identical to explicitly refitting on $N-1$ species each time, at a fraction of the computational cost.

5 Model Comparison

RMSE for species-level models: LM, Elastic Net, Random Forest, and PIP. Non-phylogenetic models use 10-fold CV; PIP uses LOOCV. In both cases each prediction excludes the observation being predicted.

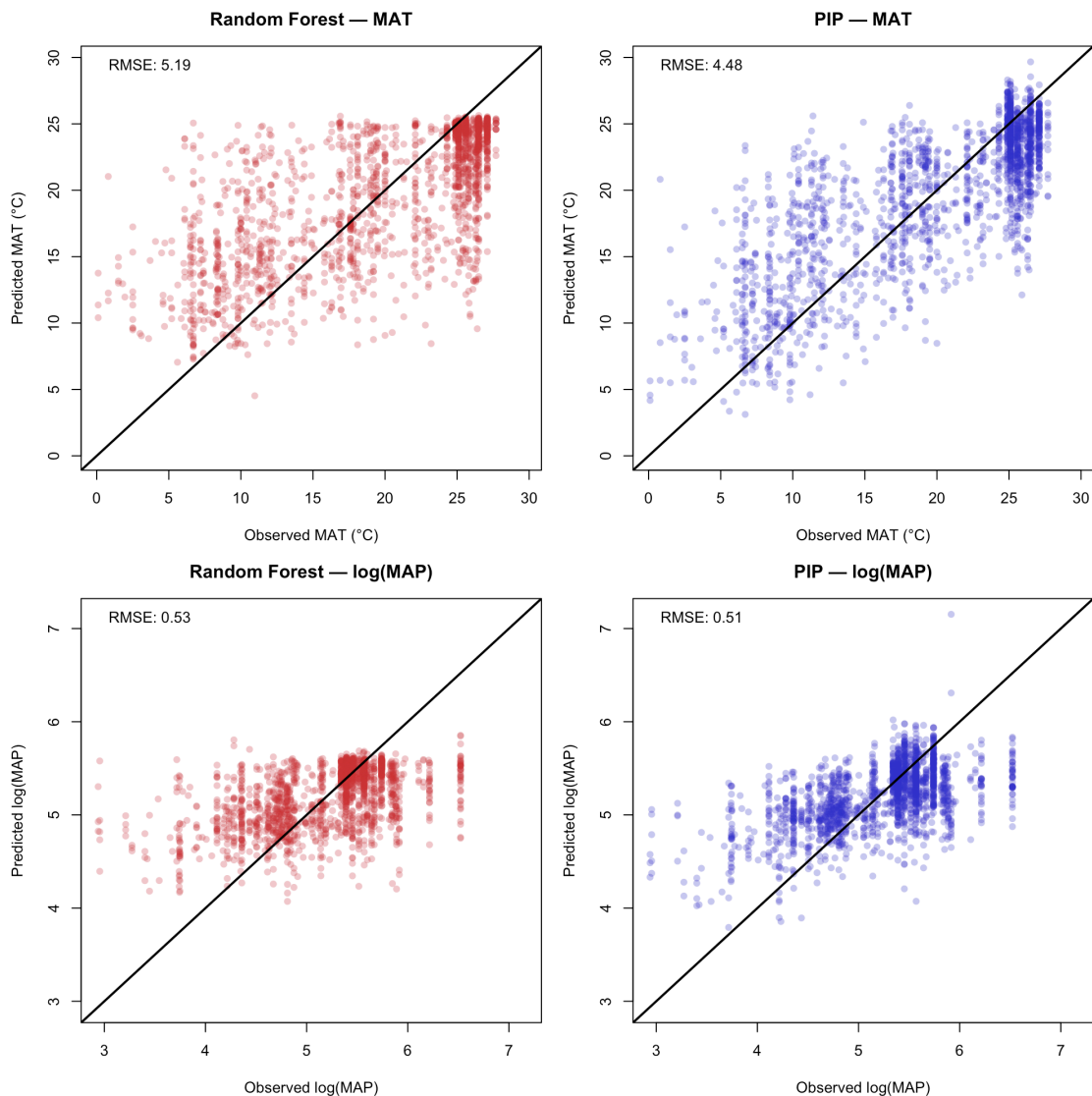
Table 6: MAT RMSE ($^{\circ}\text{C}$). LOOCV for PIP; 10-fold CV for other models.

Model	RMSE
Random Forest (species)	5.182
Elastic Net (species)	5.476
LM (species)	5.506

Table 7: log(MAP) RMSE. LOOCV for PIP; 10-fold CV for other models.

Model	RMSE
Random Forest (species)	0.531
Elastic Net (species)	0.558
LM (species)	0.577

Observed vs. predicted scatter plots (RF via 10-fold CV; PIP via LOOCV):



6 Site-Level Models

Traditional leaf physiognomy methods (CLAMP, DiLP) work at the site scale: all leaf specimens from a site are averaged to a single set of trait values, from which one climate estimate is made. The site-level models here apply the same three non-phylogenetic methods from Section 3 to site-mean trait values. Site-aggregated PIP averages species-level LOOCV predictions across all species present at each site, weighted by specimen count so that common species contribute proportionally more – consistent with how the LM uses trait means. An alternative would be to weight each species equally (presence/absence), which would treat each taxon as an independent signal regardless of how many specimens were collected.

6.1 Cross-validation results

Table 8: MAT RMSE ($^{\circ}\text{C}$) for site-level models. LOOCV for PIP; 10-fold CV for other models.

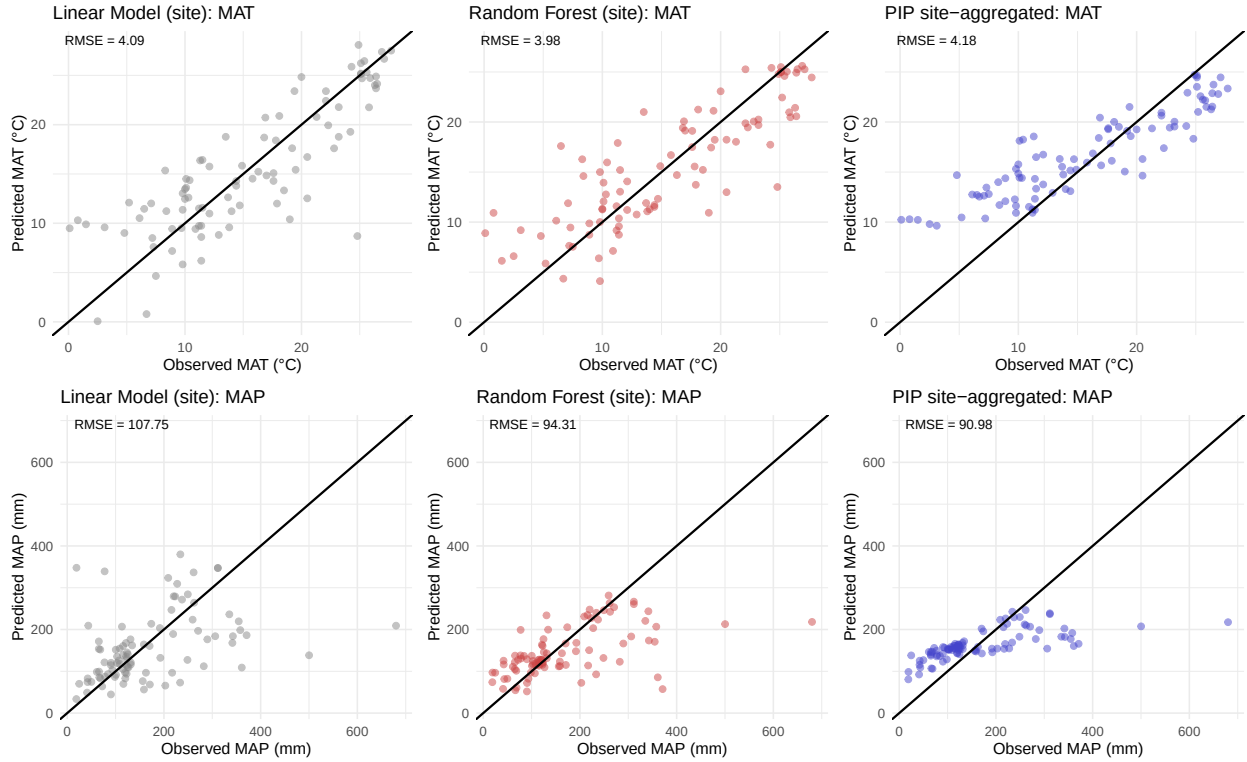
Model	RMSE
Elastic Net (site)	3.739
Random Forest (site)	3.862
LM (site)	3.931
PIP (site-aggregated)	4.184

Table 9: log(MAP) RMSE for site-level models. LOOCV for PIP; 10-fold CV for other models.

Model	RMSE
Random Forest (site)	0.529
PIP (site-aggregated)	0.555
Elastic Net (site)	0.565
LM (site)	0.594

6.2 Observed vs. predicted at the site scale

Each point is one of the 92 training sites.



For MAT, Site-aggregated PIP (4.18°C) is the weakest of the three. Random Forest (3.98°C) is one of the strongest non-phylogenetic site model. The linear model (4.09°C) falls between them. For MAP, the linear model is the weakest; Random Forest and site-aggregated PIP are similar.

7 Cross-Level Comparison and Sample Size

7.1 Full model comparison

The table below combines all models across both levels.

Table 10: MAT RMSE ($^{\circ}\text{C}$). LOOCV for PIP models; 10-fold CV for all other models.

Model	RMSE	Level
Elastic Net (site)	3.739	Site
Random Forest (site)	3.862	Site
LM (site)	3.931	Site
PIP (site-aggregated)	4.184	Site + Phylogenetic
PIP (species)	4.478	Phylogenetic
Random Forest (species)	5.182	Species
Elastic Net (species)	5.476	Species
LM (species)	5.506	Species

Table 11: $\log(\text{MAP})$ RMSE. LOOCV for PIP models; 10-fold CV for all other models.

Model	RMSE	Level
PIP (species)	0.512	Phylogenetic
Random Forest (site)	0.529	Site
Random Forest (species)	0.531	Species
PIP (site-aggregated)	0.555	Site + Phylogenetic
Elastic Net (species)	0.558	Species
Elastic Net (site)	0.565	Site
LM (species)	0.577	Species
LM (site)	0.594	Site

For MAT, site-aggregated PIP (4.18°C) improves over species-level PIP (4.48°C). Averaging within sites reduces specimen-level noise while retaining the phylogenetic adjustment. Site-level non-phylogenetic models still have lower RMSE, likely because site averaging also reduces between-species trait variation that the phylogenetic model has not fully accounted for.

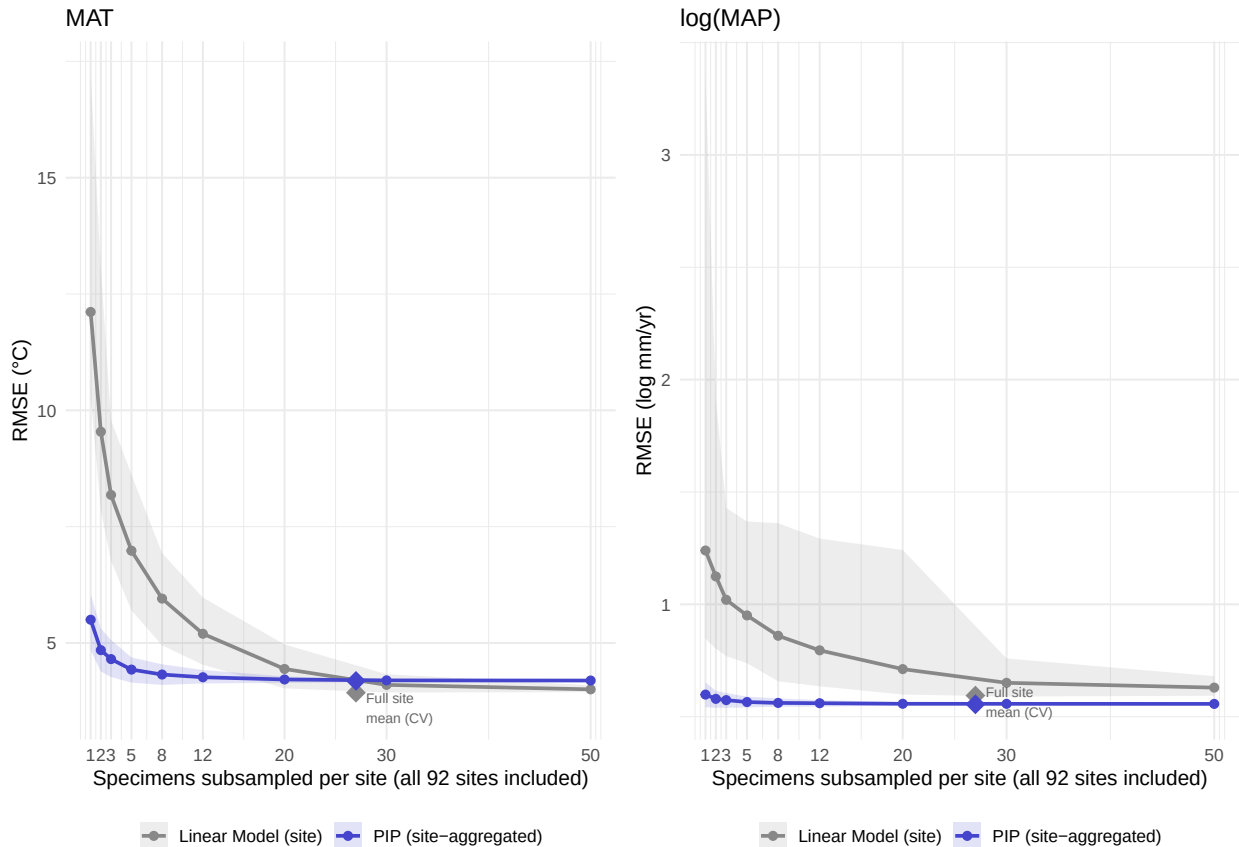
For $\log(\text{MAP})$, species-level PIP has the lowest RMSE.

Site-aggregated PIP outperforms treating each species in isolation (species-level PIP) for MAT, by gaining the noise-reduction benefit of averaging across taxa. It does not require computing site-mean traits, making it applicable when only a small number of species are available.

7.2 Sample size analysis

The comparison above uses full site means, with all specimens averaged together. In practice, a site may yield only a small number of identifiable taxa. The question is: at what sample size does each method break down, and does PIP retain an advantage when data are sparse?

To test this, we repeatedly subsample up to N specimens from each training site and compare RMSE from the site-level linear model against site-aggregated PIP. Sites with fewer than N specimens contribute all their specimens, so all 92 sites are included at every sample size. For the linear model, the subsampled specimens are averaged to a single set of trait means, which are fed to the trained LM. For PIP, the species in the subsample that are present in the training set are identified, and their species-level LOOCV predictions are averaged weighted by specimen count in the subsample, matching the LM's implicit abundance-weighting. Both models are evaluated on the same sites within each replicate: sites where no training species appeared in the subsample are excluded from both RMSE calculations. Each sample size is repeated 200 times per site.



Filled circles: bootstrap RMSE (200 replicates). Sites with fewer than N specimens contribute all their specimens. Diamonds: full site-mean CV performance (10-fold CV for LM, LOOCV for PIP), placed at the median site size ($N = 27$).

Two different evaluations are shown. The filled circles with ribbons come from the bootstrap: the LM is given trait means from a random subsample of N specimens, while PIP averages species-level LOOCV predictions for whatever species appear in those N specimens. The diamond markers show each method's full-site performance (10-fold CV for LM, LOOCV for PIP), placed at the median specimens per site ($N = 27$).

For MAT, the bootstrap shows PIP outperforming the LM at every subsampled N . This is partly a methodological asymmetry: the LM was trained on full site means and receives noisier inputs when only $N < \text{full-}N$ specimens are available, while PIP's precomputed species-level predictions do not degrade in the same way. The diamond markers resolve this: at the full site mean (proper CV), the LM (3.931°C) outperforms site-aggregated PIP (4.184°C). The LM curve is converging toward its CV performance, and the crossover occurs somewhere around the median site size.

For log(MAP), PIP holds its advantage even at the full site mean, consistent with its lead in the main comparison table.

For sites with fewer than roughly 27 specimens (the median site size in the training data), PIP is expected to outperform the site-level LM for MAT. For larger, well-sampled sites approaching the full site-mean condition, the LM is competitive. For MAP, PIP is preferred at any sample size.

8 Fossil Prediction

8.1 Phylogenetic placement

Fossils are grafted onto a scaffold phylogeny (`tre_scaffold.tre`). The scaffold contains a family-level angiosperm backbone (two crown tips per family across all 515 WCVF angiosperm families, approximately 943 tips) with all 1,847 training species added. This ensures that any angiosperm family can be placed in the tree, including families not represented in the training data.

For each fossil, placement proceeds through a fallback sequence:

1. **Genus:** if tips from the fossil’s genus are present, attach at their MRCA
2. **Family:** if the genus is absent, find all genera in the family via the full WCVF taxonomy and attach at their MRCA
3. **Order:** as above at the order level
4. **Root:** if the order is not in the WCVF backbone, attach at the root; the phylogenetic adjustment will be near zero in this case

The edge length ℓ is set so the fossil tip terminates at its stated stratigraphic age:

$$\ell = (h - t_{\text{fossil}}) - \text{height}(\text{node})$$

where h is the crown age of the tree and t_{fossil} is the fossil’s age in Ma. If a fossil’s stated age is older than its placement node, the code walks up the tree to find the branch that spans the fossil’s age and splits it there, placing the tip at the correct time depth.

8.2 Trait imputation

Fossils often have incomplete trait data. Missing trait values are imputed using a `bagImpute` model trained on extant species using traits only, not the climate response. The climate response is unknown for fossils and cannot be used as a predictor in imputation. Any trait column absent from the fossil input is padded with NA before imputation.

The traits used for imputation are those in the active predictor set for each target, as selected by Elastic Net.

8.3 Applying PIP to fossils

After placement, the cross-covariance $\mathbf{V}[\text{train}, \text{fossil}]$ is extracted from the pruned combined tree and the PIP formula is applied as described in Section 4.3. The training VCV $\mathbf{V}[\text{train}, \text{train}]$ is reused from the extant-species fitting. This is valid because the covariance between any two training species equals the height of their MRCA, which does not change when additional taxa are added to the tree.

8.4 Results

Table 12: Fossil climate predictions (test specimens). CIs reflect coefficient uncertainty only. See Section 4.4 for a description of excluded uncertainty components.

Specimen	MAT (°C)	95% CI	MAP (mm)	95% CI
Quercus_miocene_1	16.7	[15, 18.4]	136	[118, 157]
Acer_oligocene_1	21.3	[19.8, 22.9]	161	[140, 185]
Ficus_eocene_1	23.4	[21.8, 25]	297	[257, 343]

These are test specimens with synthetic trait data and should not be interpreted as palaeoclimate estimates. The table shows the output format for real fossil inputs.

8.5 Input format

Fossil specimens are supplied via `data/fossil_traits.csv`. Required columns:

Column	Description
<code>fossil_name</code>	Unique specimen identifier, used as the tip label in the tree
<code>genus</code>	Genus name, for placement
<code>family</code>	Family name, fallback placement
<code>order</code>	Order name, fallback placement
<code>age_ma</code>	Fossil age in millions of years before present
<i>(trait columns)</i>	Any subset of the 12 fossil-measurable traits; missing values are imputed

Additional columns are ignored.

9 Software and Reproducibility

All analyses are implemented in R. The PGLS fitting and PIP formula follow the source code of Gardner, Baker, Venditti & Organ (2024; *Nature Communications*), originally from Freckleton (2015). The dated angiosperm phylogeny is from the WCVF backbone tree.

Key packages: `ape`, `phytools`, `caret`, `glmnet`, `ranger`, `pdp`, `ggplot2`.

Scripts run in order:

```
00_data_cleaning.R      # trait filling, species + site aggregation, scaffold tree
01_nophy_regression.R   # LM, Elastic Net, Random Forest (species and site level)
02_phy_regression.R     # PGLS (MAT + MAP), save PIP components
03_comparison.R         # LOOCV, RMSE tables, importance, plots
04_fossil_predictions.R # fossil climate prediction via PIP
```